Technical Report: Conditional GAN Synthesis and Evaluation on FashionMNIST

1. Introduction to Generative Adversarial Networks (GANs)

The Adversarial Objective

Generative Adversarial Networks (GANs) are formulated as a zero-sum minimax game between two deep neural networks: the Generator (G) and the Discriminator (D). The Generator seeks to map a latent noise distribution p_z(z) to the data space, while the Discriminator attempts to distinguish between samples from the true data distribution p_{data}(x) and synthetic samples produced by G. Formally, this relationship is defined by the objective function: (MinMaxLoss) The optimization utilizes binary cross-entropy (BCE) to drive the competition, theoretically reaching a Nash equilibrium where G recovers the underlying data manifold and D is unable to differentiate between real and fake samples better than random chance.

Conditional GAN Framework

This implementation extends the vanilla framework into a Conditional GAN (cGAN), where both G and D are conditioned on class labels y . By incorporating categorical information via nn.Embedding, the model learns the conditional distribution P(x|y) . This enables targeted synthesis of the 10 specific FashionMNIST categories, transforming the generation process from a stochastic mapping to a controlled semantic projection.

2. Model Architecture and Design Decisions

The Generator (G)

The Generator is architected to upsample a joint representation of noise and class identity into a 28 X 28 grayscale image. The transition from low-dimensional latent space to high-dimensional pixel space is achieved through the following layers:

Layer Type	Input Dim	Output Dim	Kernel	Stride	Padding	Activation/Norm
Embedding	10	10	-	-	-	N/A
Fully Connected	$z_{dim} + 10$	$256 \times 4 \times 4$	-	-	-	BatchNorm, ReLU
ConvTranspose2d	$256 \times 4 \times 4$	$128 \times 7 \times 7$	4×4	1	0	BatchNorm, ReLU
ConvTranspose2d	$128 \times 7 \times 7$	$64 \times 14 \times 14$	4×4	2	1	BatchNorm, ReLU
ConvTranspose2d	$64 \times 14 \times 14$	$1 \times 28 \times 28$	4×4	2	1	Tanh

The choice of Tanh as the final activation ensures the output pixel values are normalized to the (-1, 1) range, matching the pre-processing applied to the ground truth dataset.

The Discriminator (D)

The Discriminator employs a series of strided convolutions to perform binary classification. Pedagogically, this architecture prioritizes stability over raw capacity.

1. **Input Processing:** To condition the Discriminator, class labels are embedded and projected via a linear layer to a 1X28X28 feature map. This map is concatenated along the channel dimension with the input image, resulting in a 2-channel input (Image + Label Projection).
2. **Convolutional Progression:**
3. nn.Conv2d(2, 64, 4, 2, 1): Reduces spatial dimensions to 14 X 14.
4. nn.Conv2d(64, 128, 4, 2, 1): Reduces spatial dimensions to 7 X 7 .
5. nn.Conv2d(128, 256, 3, 2, 1): Reduces spatial dimensions to 4 X 4 .
6. nn.Conv2d(256, 1, 4, 1, 0): Collapses features to a single scalar logit.
7. **Spectral Normalization:** Every convolutional layer is wrapped in spectral_norm. This technique constrains the spectral norm of the weight matrices to satisfy a Lipschitz constant K<=1 . Unlike weight clipping, spectral normalization provides a more graceful constraint on gradient magnitudes, preventing the Discriminator from becoming too sharp (which often leads to vanishing gradients for the Generator).
8. **Numerical Stability:** The final layer omits a Sigmoid activation, feeding directly into **BCEWithLogitsLoss**. This approach is numerically superior as it leverages the log-sum-exp trick, preventing overflow/underflow issues during backpropagation.

3. Training Methodology and Stability Optimization

Hyperparameters

The model was trained for 100 epochs using the following parameters:

- **Batch Size:** 128
- **Optimizer:** Adam (beta_1 = 0.5, beta_2 = 0.999)
- **Learning Rate:** 0.002

Stability Techniques

Maintaining adversarial equilibrium requires specific heuristics to prevent the Discriminator from overpowering the Generator:

1. **Label Smoothing:** Using the smooth_real_labels function, the target for real images is sampled from a uniform distribution [0.9, 1.0] . This softens the penalty for the Discriminator, preventing it from reaching absolute certainty and ensuring a continuous gradient signal for G .
2. **Heuristic-Based Update Scheduling:** The Discriminator is only updated if loss_D.item() > 0.3. This conditional logic ensures that if the Discriminator is already highly proficient, the update cycle is skipped to allow the Generator to catch up, maintaining the competitive balance necessary for learning.

4. Empirical Results and Training Dynamics

Loss Curve Analysis

The training dynamics demonstrate the classic oscillatory nature of GANs. Early epochs show sharp fluctuations as the networks establish their respective roles. Post-initialization, the system stabilizes: the Discriminator loss fluctuates within the $1.3–1.4$ range, and the Generator loss settles between $0.7–0.8$. This convergence suggests a healthy equilibrium where neither player dominates the game.

Visual Inspection and Mode Collapse Assessment

Temporal analysis of generated samples reveals a progression from unstructured noise at Epoch 0 to high-fidelity garments by Epoch 95. The model shows no signs of **Mode Collapse**. Visual inspection of the 10-class grids demonstrates significant intra-class variation (e.g., varying sleeve lengths in T-shirts and distinct boot heights) and sharp inter-class boundaries. The model maintains the full diversity of the FashionMNIST distribution without collapsing to "safe" or repetitive samples.

5. Quantitative Evaluation: FID Scores

The Fréchet Inception Distance (FID) provides an objective measure of quality by comparing the statistics of activations from a pre-trained Inception-v3 network for both real and generated sets.

Training Checkpoint	FID Score
Epoch 10	92.32
Epoch 50	31.54
Epoch 100	**25.45**

The consistent reduction in FID confirms that the architectural constraints (Spectral Norm, Label Smoothing) successfully supported long-term learning, with the best score of 25.45 achieved at the final checkpoint.

6. Latent Space Interpretation

Spherical Linear Interpolation (Slerp)

To explore the latent manifold, we utilize Spherical Linear Interpolation (Slerp) between two random vectors z_1 and z_2. Slerp is mathematically superior to standard linear interpolation (Lerp) for high-dimensional Gaussian latent spaces; because the mass of a high-dimensional Gaussian is concentrated on a hypersphere (the "soap bubble" effect), Lerp would cut across the center of the sphere into a low-probability region, causing "hollow" artifacts. Slerp preserves the vector magnitude, maintaining the manifold's curvature. Transitions across Labels 0, 1, and 2 show smooth variations in style while the semantic identity remains rigidly defined by the class label.

Class Morphing Analysis

By fixing the noise vector z and interpolating between the embeddings of Class 0 (T-shirt) and Class 1 (Trouser), we observe the following:

- **Smooth Transitions:** The model demonstrates a continuous morphing of geometry (e.g., sleeves shrinking while the torso elongates).
- **Disentanglement:** The latent vector z appears to control global "style" or "pose," while the embedding controls categorical "content." This separation of concerns indicates a well-disentangled latent representation.
- **Semantic Manifold:** The embedding space is confirmed to be semantically continuous rather than a collection of discrete, isolated points.
- **Absence of Artifacts:** The lack of sharp visual glitches during morphing suggests the Generator has learned a robust mapping of the entire intermediate embedding space.

7. Conclusion

The implemented Conditional GAN architecture successfully synthesized high-quality, diverse samples across all 10 FashionMNIST categories. The integration of spectral normalization and heuristic-based update scheduling proved critical in maintaining training stability, ultimately resulting in a competitive FID score of 25.45. The latent space analysis further validates that the model has learned a continuous, semantically meaningful manifold capable of both style variation and class morphing.